Attention with Linear Biases for Input Length Extrapolation (Press et al. 2021)

Injecting non-learned, head-specific static negative linear penalties into query-key attention scores (ALiBi) enables Transformer language models to extrapolate smoothly to sequences far longer than those observed during training.

GRADE: HIGH | STATUS: STABLE | DOMAIN: TECHNICAL, ORGANIZATIONAL, PHILOSOPHICAL | REPLICATION: VERIFIED

WHAT WE KNOW (Narrowest Supported Conclusion)

Replacing learned or sinusoidal position embeddings with static, linearly decaying attention biases proportional to token distance (softmax(q k^T - m · (i - j))) allows autoregressive language models trained on short sequences (e.g., 512 or 1024 tokens) to extrapolate stably to sequences several times longer (e.g., 2048 to 8192+ tokens) without perplexity explosion.

EVIDENCE AND CAUSAL ASSESSMENT

Controlled comparisons against sinusoidal positional encodings, learned positional embeddings, T5 relative bias, and rotary position embeddings prove that linear distance-proportional penalties prevent attention score saturation on unseen sequence lengths.

EFFECT SIZE AND PERSISTENCE

Models trained on sequences of length 512 with ALiBi maintained declining or flat validation perplexity when evaluated on sequences up to 2,048 tokens (+4× extrapolation), whereas models with sinusoidal or learned embeddings suffered catastrophic perplexity spikes within 50–100 tokens past the training horizon.

BIAS, HETEROGENEITY, AND CONFOUNDING

Classified as: robust (<20%).

Classified as robust (<20%). While ALiBi guarantees stable perplexity extrapolation, its strong monotonic recency bias can impair long-range associative recall and needle-in-a-haystack retrieval tasks at extreme sequence lengths (>32K tokens).

WHAT THIS DOES NOT MEAN (Strict Boundary Conditions)

Length extrapolation does not mean a model retains uniform recall over distant tokens; linear decay guarantees mathematical stability rather than lossless associative memory.

WHAT ACTION IS REASONABLY JUSTIFIED

Use relative positional encodings such as ALiBi or RoPE with frequency scaling when designing language models intended for context-length expansion beyond their pre-training window.

VERIFIED SOURCES & DOI IDENTIFIERS

- Press, O., Smith, N. A., & Lewis, M. (2021). Train Short, Test Long: Attention with Linear Biases Enables Input Length Extrapolation. ICLR 2022. https://arxiv.org/abs/2108.12409